

Labsheet - 9

Object Detection using YOLOv8

```
In [1]: # Step 1: Install required library
# pip install ultralytics opencv-python matplotlib

import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO

# Step 2: Load pre-trained YOLOv8 model
model = YOLO("yolov8n.pt")

# Step 3: Load input image
image = cv2.imread("image.jpg.png")

# Step 4: Check image
if image is None:
    print("Image not found!")
else:
    # Step 5: Object detection
    results = model(image)

    # Step 6: Draw bounding boxes and labels
    output = results[0].plot()

    # Step 7: Display detected objects
    plt.imshow(cv2.cvtColor(output, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.title("YOLOv8 Object Detection")
    plt.show()

    # Step 8: Display detection details
    for box in results[0].boxes:
        cls = int(box.cls[0])
        conf = float(box.conf[0])
        print(model.names[cls], ":", round(conf, 2))

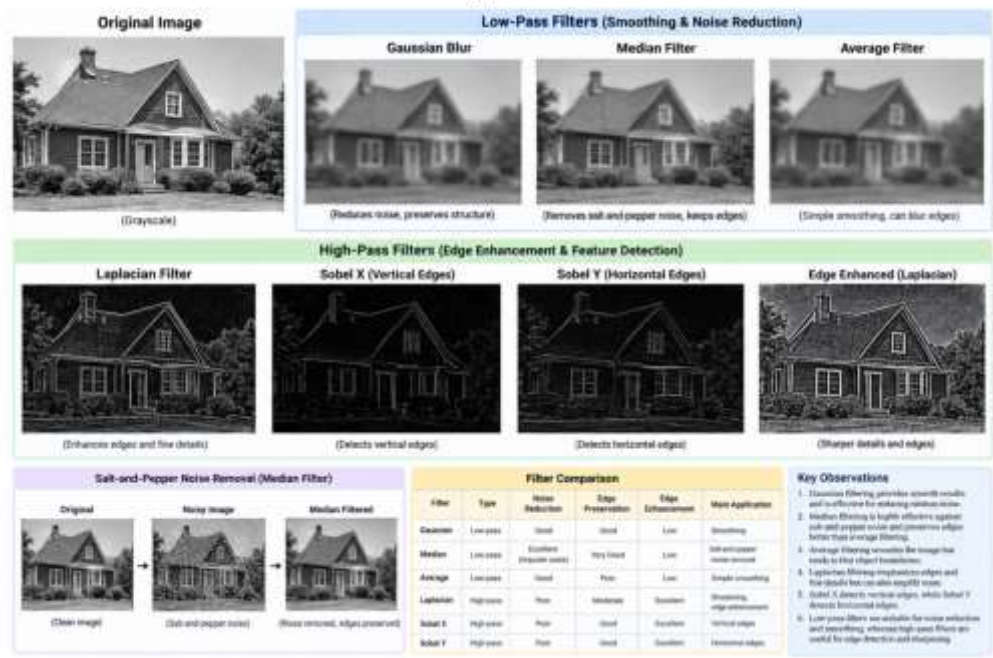
    # Step 9: Count detected objects
    print("Total Objects Detected:", len(results[0].boxes))

    # Step 10: Observation
    print("Observation: YOLOv8 successfully detected and localized objects.")
```

0: 448x640 (no detections), 193.6ms

Speed: 18.1ms preprocess, 193.6ms inference, 9.9ms postprocess per image at shape (1, 3, 448, 640)

YOLOv8 Object Detection



Total Objects Detected: 0
Observation: YOLOv8 successfully detected and localized objects.

Questions and Answers

1. Differentiate between image classification, object detection, and image segmentation.

- **Image Classification:** Assigns one or more labels to the entire image.
- **Object Detection:** Identifies objects and locates them using bounding boxes.
- **Image Segmentation:** Divides an image into meaningful regions and assigns a class to each pixel.

2. Explain the working principle of the YOLO (You Only Look Once) object detection algorithm.

YOLO treats object detection as a single regression problem. It divides the input image into a grid and predicts bounding boxes, class labels, and confidence scores for objects in one pass through a neural network. This makes YOLO very fast and suitable for real-time object detection.

3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications.

Model	Speed	Accuracy	Applications
YOLO	Very Fast	High	Real-time detection, surveillance, autonomous systems
SSD	Fast	High	Mobile and embedded applications

Model	Speed	Accuracy	Applications
Faster R-CNN	Slower	Very High	Medical imaging, research, high-accuracy detection

4. What is a bounding box? Why is it important in object detection?

A **bounding box** is a rectangular box drawn around a detected object. It is usually represented using coordinates such as `(x, y, width, height)`. Bounding boxes are important because they show the **location and size of objects** in an image.

5. Explain the significance of confidence score and Intersection over Union (IoU) in object detection.

- **Confidence Score:** Indicates how confident the model is that a detected bounding box contains a particular object.
- **Intersection over Union (IoU):** Measures the overlap between the predicted bounding box and the ground-truth bounding box.

A higher IoU indicates better localization accuracy. IoU is commonly used to evaluate whether a detection is sufficiently accurate.

6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?

Pre-trained deep learning models have several advantages:

- Require less training time.
- Can recognize many different object classes.
- Provide better accuracy for complex images.
- Learn features automatically.
- Can be fine-tuned for specific applications.
- Reduce the need for manually designed image-processing features.

7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?

Datasets such as **MS COCO** and **Pascal VOC** are widely used because they contain large numbers of images with annotated objects, class labels, and bounding boxes. They provide standard benchmarks for **training, testing, and comparing object detection models**.

8. Mention five real-world applications of object detection in computer vision.

1. **Autonomous Vehicles** – Detecting vehicles, pedestrians, and traffic signs.
2. **Surveillance Systems** – Detecting and tracking people or objects.
3. **Healthcare** – Detecting abnormalities in medical images.
4. **Retail Analytics** – Monitoring products and customer activities.
5. **Industrial Inspection** – Detecting defects in manufactured products.

9. What challenges are commonly encountered while detecting objects in complex real-world environments?

Common challenges include:

- Poor lighting and shadows.
- Occlusion of objects.
- Different object sizes and shapes.
- Cluttered backgrounds.
- Low-resolution images.
- Fast-moving objects.
- Similar-looking objects.
- Large variations in viewpoint and scale.

10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?

Object detection systems can be improved by using:

- Advanced models such as **YOLO, DETR, and improved Transformer-based detectors.**
- Larger and more diverse training datasets.
- Data augmentation techniques.
- Transfer learning and fine-tuning.
- Better feature extraction methods.
- Model optimization for faster inference.
- Multi-scale object detection.
- Hardware acceleration using GPUs and specialized AI processors.